A halting configuration of a Turing machine entails the halting sentence

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openlogic:turing-machines/undecidability:local-snapshot-20261013;
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Editorial note. The counted statement is the author's own lemma environment lem:halt-config-implies-halt (source lines 1648-1651, the \ollabel on line 1649); the proof environment begins two lines below it and is its own proof, closing at source line 1673.

Editorial note. Neither the statement nor the proof contains an \olref or a \cite, so no source pointer and no bibliography entry are needed; the objects C(M,w,n), E(M,w) and the relation delta are all introduced by the author inside the retained unit, and every symbol used is declared in the retained preamble.

Editorial note. The retained preamble is the minimal working preamble shipped with the frozen OLP unit (OLP's own style stack needs prooftree, ntheorem, extarrows, mathdots, isomath and subfiles, which the project toolchain does not have). Author text is untouched; only the packaging is the intake line's.

Editorial note. Section and theorem counters are reset to zero so the retained lemma prints with the numbering of the retained front matter rather than continuing the frozen file's count.

Editorial scope. One original result of the Open Logic Project's undecidability unit, counted once: a halting configuration entails the author's halting sentence.

Allowed standard prerequisites (editorial).

- The sentence C(M,w,n) describing a configuration. C(M,w,n) is the first-order sentence whose clauses record the state of M at time n, the square being scanned, and the contents of the finitely many relevant tape squares at time n, as defined in the author's definition immediately before the counted lemma.. The proof reads the clauses Q_q(m,n) and S_sigma(m,n) out of C(M,w,n) at source lines 1661-1663.
- The sentence E(M,w) expressing that M halts. E(M,w) is the existential sentence saying that there are a time and a square at which M is in a state q scanning a symbol sigma with delta(q,sigma) undefined.. The proof exhibits the pair (q,sigma) as one of the disjuncts of E(M,w) at source lines 1665-1672.

Lemma 0.1. *If M run on input w is in a halting configuration after n steps, then $!C(M, w, n) \models !E(M, w)$.*

Proof. Suppose that M halts for input w after n steps. There is some state q, square m, and symbol σ such that:

1. After n steps, M is in state q scanning square m on which σ appears.
2. The transition function $\delta(q, \sigma)$ is undefined.

$!C(M, w, n)$ is the description of this configuration and will include the clauses $Q_q(\underline{m}, \underline{n})$ and $S_\sigma(\underline{m}, \underline{n})$. These clauses together imply $!E(M, w)$:

$$\exists[x][\exists[y][(\bigvee_{\langle q,\sigma\rangle \in X} (Q_q(x, y) \wedge S_\sigma(x, y)))]]$$

since $Q_{q'}(\underline{m}, \underline{n}) \wedge S_{\sigma'}(\underline{m}, \underline{n}) \models \bigvee_{\langle q,\sigma\rangle \in X}(Q_q(\underline{m}, \underline{n}) \wedge S_\sigma(\underline{m}, \underline{n}))$, as $\langle q', \sigma'\rangle \in X$. □